

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

ANALOG AND MIXED SIGNAL DESIGN [ECE-4061]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) An IC Fab can fabricate MOS transistors with $W/L=8\mu\text{m}/0.8\mu\text{m}$. The other process technology parameters that the Fab offers are the following: Oxide thickness $t_{ox}=8\text{nm}$, Electron mobility $\mu_n=350\text{cm}^2/\text{V}\cdot\text{s}$ and the threshold Voltage $V_{th}=0.8\text{V}$. Calculate C_{ox} and K_n' . Also find drain current for $V_{gs}=3\text{V}$. (4)
 - A)
 - B) Explain the advantageous of using MOSFET as a load. (3)
 - C) Explain how multifinger transistors are used to avoid folding of MOSFETs. (3)
- 2) For the given circuit in figure 2A assume Q1 and Q2 are perfectly matched and have the device parameters channel length = $1\mu\text{m}$, channel width $10\mu\text{m}$ $V_{th}=0.7\text{V}$ $\mu_n C_{ox}=200\mu\text{A}/\text{V}^2$ and $\lambda=0$. Find the value of R to get $100\mu\text{A}$ output current, what is the lowest possible value of V_{out} .
 - A)

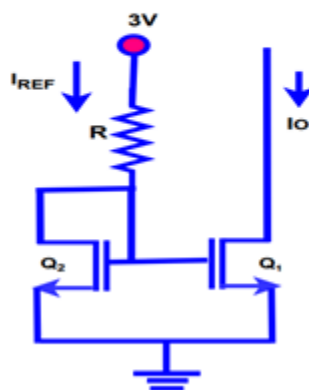


Figure 2A

- B) (3)

For the given common source amplifier in figure 2B. find the transconductance of the MOSFET. Assume that at the operating signal frequency, the capacitor act as short circuit. The following are the MOSFET parameters, $V_T=1.2V$, $\mu_n C_{ox}(W/L)=0.75mA/V^2$. Assume $R_F=20M\Omega$

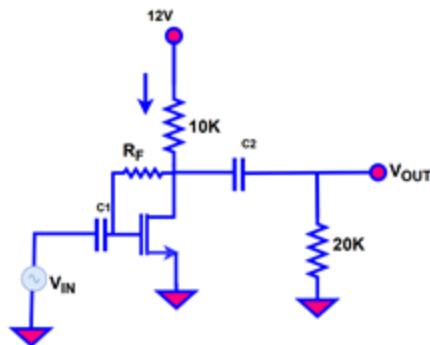


Figure 2B

C)

For the given common gate amplifier in figure 2C, the voltage gain is 10 and input impedance is 150Ω . The power supplied by the 4.5V DC voltage source is 10mW. Calculate the transconductance of the transistor. Assume that at the operating signal frequency, the capacitor act as short circuit. The following are the MOSFET parameters, $V_T=0.7V$, $\mu_n C_{ox}(W/L)=15mA/V^2$ and $\lambda=0$. Assume $R_1=1M\Omega$, $R_2=0.5M\Omega$.

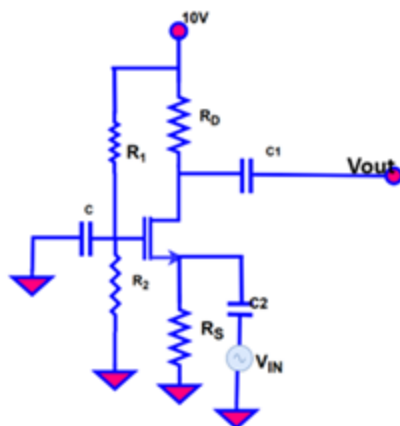


Figure 2C

(3)

- 3) Design an NMOS differential pair for a voltage gain of 5 and power budget of 2mW subject to the condition that the stage following the differential pair requires an input CM level of at least 1.6V. Assume $M_n C_{ox}=100\mu A/V^2$, $\lambda=0$ and $V_{dd}=1.8V$. What is the minimum allowable input CM level if $V_{th}=0.4V$.

A)

(4)

B)

(3)

In the MOS differential pair of figure 3B, $V_{CM} = 1V$, $I_{SS} = 1\text{ mA}$, and $R_D = 1k\Omega$. What is the minimum allowable supply voltage if the transistors must remain in saturation? Assume $V_{TH,n} = 0.5V$.

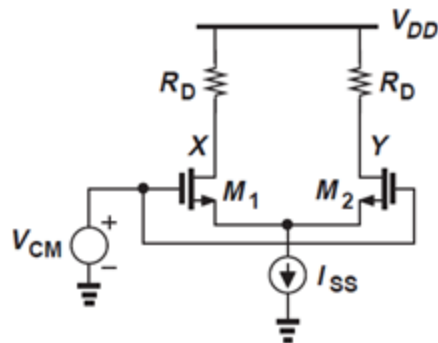


Figure 3B

- C) Derive the input/output characteristic of a MOS differential pair if (a) the tail current is halved or (b) the transistor aspect ratio is halved. (3)
- 4) Explain the working operation of OTA with neat circuit diagram. (4)
- A)
- B) Design an NMOS differential pair for a voltage gain of 10 and power budget of 2mW subject to the condition that the stage following the differential pair requires an input CM level of at least 1.2V. Assume $M_n C_{ox} = 50\mu A/V^2$, $\lambda = 0$ and $V_{DD} = 2V$. (3)
- C) Derive the input/output characteristic of a MOS differential pair if (a) the tail current is halved (1/2th) and (b) the transistor aspect ratio is one fourth (1/4th). (3)
- 5) Design an 4bit SAR ADC having DAC reference voltage of 16V, and find the respective digital word for $V_{in} = 10V$, $V_{in} = 12V$. (4)
- A)
- B) Design a 3bit R-2R ladder DAC, find the respective output voltage for all combinations of input digital word, $V_{ref} = 5V$, $R = 1K\Omega$ and $R_F = 10K\Omega$. (3)
- C) Design a 3-bit charge-scaling DAC and find the value of the output voltage for $D_2 D_1 D_0 = 010$ and 101 . Assume that $V_{ref} = 5V$ and $C = 0.5\text{ pF}$. (3)

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